203740: porphyritic metasyenite, Wallaby mine

(Yilgarn Craton granites, Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Laverton (SH 51-2), Laverton (3340)
MGA Zone 51, 432246E 6807536N

Sampled on 2 September 2020

This sample was collected from the 1619.4 – 1621.4 m depth interval of drillcore WBD017UD, drilled in 2015 by GSM Mining Company Pty Ltd at the Wallaby mine (Standing, 2008). The drillhole is located on Mount Weld Station, at the northern end of Lake Carey, about 16.4 km southwest of Mount Weld homestead, 15.8 west-northwest of Brumby Well, and 11.5 km east-southeast of Mount Margaret.

Geological context

The unit sampled is a porphyritic metasyenite (Fig. 1) within the Laverton greenstone belt of the Kurnalpi Terrane (Stewart, 2001; Standing, 2008). The porphyritic metasyenite consists of multiple intrusions, mainly metamorphosed syenite, monzonite and minor carbonatite, that intruded the Wallaby Conglomerate in the Lake Carey domain (Standing, 2008). Detrital zircons in a sandstone horizon within conglomerate at the Wallaby mine yielded a conservative maximum depositional age of 2673 ± 5 Ma (sample E319, Kositcin et al., 2008). Titanite from a carbonatitic intrusion at the Wallaby mine yielded a U–Pb crystallization age of 2664 ± 3 Ma (Salier et al., 2004). A porphyritic syenite about 11.0 km to the south-southeast in drillcore LCDD0076 yielded an igneous crystallization age of 2663 ± 8 Ma (GSWA 203738, Wingate et al., 2022a). A felsic volcanic rock about 11.5 km to the south-southeast in drillcore LCDD0013 yielded an igneous crystallization age of 2696 ± 6 Ma (GSWA 203739, Wingate et al., 2022b). Detrital zircons in a polymictic breccia unit at the Granny Smith mine, about 13.3 km to the northeast, indicated a youngest detrital age component at 2663 ± 5 Ma (sample E401, Kositcin et al., 2008). A feldspar–quartz-porphyritic rock at Ida Hill, about 22 km to the northeast, provided an igneous crystallization age of c. 2660 Ma (sample E258, Kositcin et al., 2008).

Petrographic description

The sample is an altered, weakly porphyritic metasyenite (Fig. 1) containing phenocrysts of K-feldspar up to 12 mm long within a medium- to coarse-grained (mostly 5–7 mm) matrix of interlocking K-feldspar and minor plagioclase (<5%). Feldspars are partly recrystallized and exhibit sutured grain boundaries and subgrains on the margins. Alteration is dominated by coarse sericite and carbonate minerals. Minor chlorite occurs in patches and discontinuous stringers throughout the groundmass. Sparse sulfide minerals are associated with carbonate minerals and chlorite.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, and anhedral to subhedral. The crystals are up to 200 μm long, and mainly equant, with aspect ratios up to 2:1. Most zircons contain numerous inclusions of other minerals, are strongly altered, and exhibit mottled textures, although relict concentric zoning is preserved in some crystals. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed over three sessions, on 9–10, 15–16, and 22–23 October 2020, using SHRIMP-B. Analyses 1.1 to 6.1 (spot numbers 1–6) were obtained during the first session, together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.39%. Analyses 7.1 to 15.1 (spot numbers 7–16) were obtained during the second session, together with 16 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Analyses 1.2 to 27.1 (spot numbers 17–30) were obtained during the third session, together with 12 analyses of the M257 standard, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.31% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Thirty analyses were obtained from 27 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

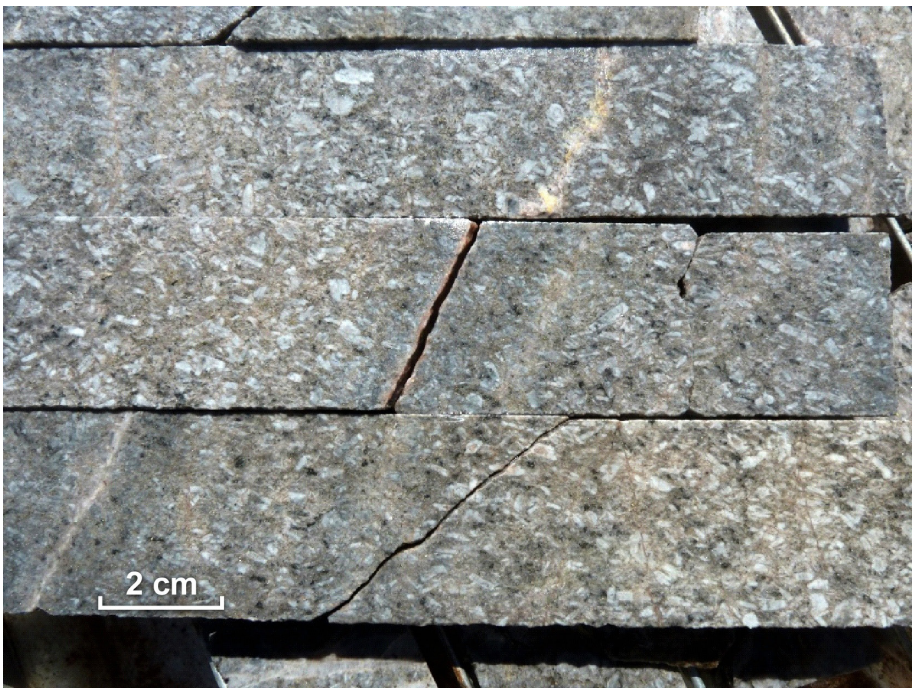

Figure 1. Drillcore image of sample 203740: porphyritic metasyenite, Wallaby mine

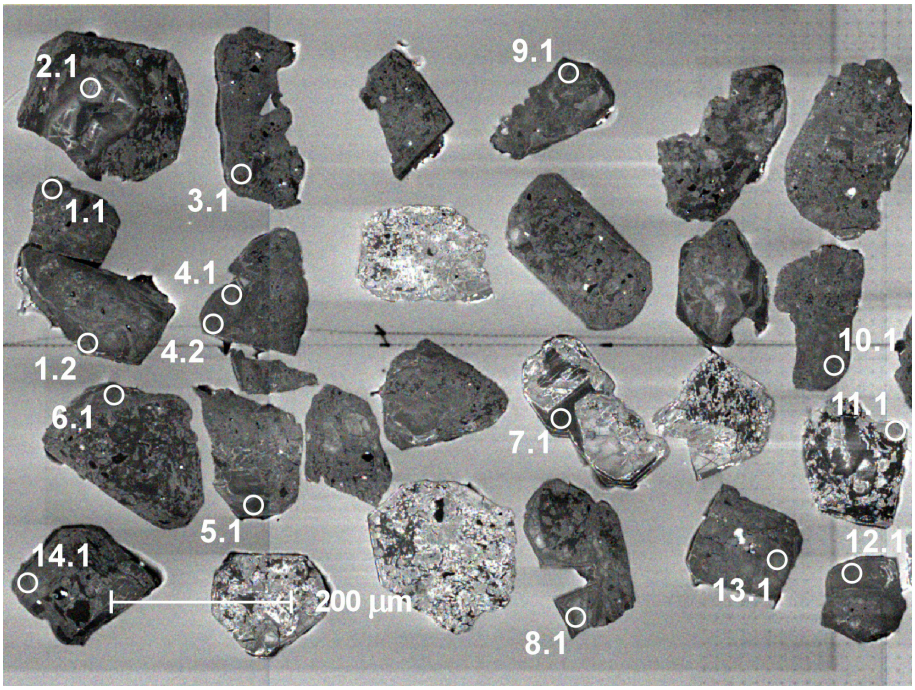

Figure 2. Cathodoluminescence image of representative zircons from sample 203740: porphyritic metasyenite, Wallaby mine. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 203740: porphyritic metasyenite, Wallaby mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	26	23.1	317	367	1.16	0.068	2.157	0.025	0.17332	0.00107	2.159	0.025	0.17272	0.00111	2454	24	2584	11	5.0
I	8	8.1	307	69	0.23	0.176	2.139	0.025	0.17662	0.00110	2.143	0.025	0.17505	0.00121	2469	24	2607	12	5.3
I	23	20.1	318	104	0.33	0.142	2.114	0.060	0.17640	0.00185	2.117	0.060	0.17513	0.00190	2494	59	2607	18	4.4
I	21	18.1	427	533	1.25	0.195	2.023	0.065	0.17728	0.00422	2.027	0.065	0.17555	0.00425	2585	68	2611	40	1.0
I	10	10.1	401	95	0.24	0.081	2.121	0.023	0.17663	0.00097	2.122	0.023	0.17591	0.00102	2489	23	2615	10	4.8
I	11	11.1	390	161	0.41	0.124	2.138	0.024	0.17829	0.00106	2.140	0.024	0.17719	0.00113	2471	23	2627	11	5.9
I	22	19.1	333	22	0.06	0.000	1.982	0.101	0.17805	0.00109	1.982	0.101	0.17805	0.00109	2634	111	2635	10	0.0
I	25	22.1	257	22	0.09	0.739	2.102	0.039	0.18522	0.00132	2.118	0.039	0.17862	0.00184	2493	39	2640	17	5.6
I	15	4.2	350	33	0.10	0.032	2.021	0.024	0.17920	0.00109	2.021	0.024	0.17891	0.00111	2591	25	2643	10	2.0
I	12	12.1	373	107	0.29	0.045	2.054	0.023	0.17971	0.00105	2.055	0.023	0.17931	0.00107	2556	24	2647	10	3.4
I	29	26.1	242	45	0.19	0.043	2.009	0.025	0.18003	0.00124	2.010	0.025	0.17965	0.00127	2603	27	2650	12	1.8
I	6	6.1	292	64	0.22	0.072	2.033	0.022	0.18029	0.00127	2.034	0.022	0.17965	0.00130	2578	23	2650	12	2.7
I	19	16.1	179	17	0.09	0.000	1.980	0.027	0.18121	0.00140	1.980	0.027	0.18121	0.00140	2636	30	2664	13	1.1
I	1	1.1	398	41	0.10	-0.032	1.967	0.020	0.18207	0.00118	1.967	0.020	0.18236	0.00119	2650	23	2674	11	0.9
I	4	4.1	233	31	0.13	0.020	1.996	0.024	0.18271	0.00142	1.997	0.024	0.18254	0.00143	2618	26	2676	13	2.2
P	2	2.1	1004	3552	3.54	0.113	2.632	0.024	0.13768	0.00093	2.635	0.024	0.13668	0.00096	2074	16	2185	12	5.1
P	17	1.2	651	102	0.16	0.015	2.291	0.031	0.15790	0.00067	2.291	0.031	0.15777	0.00068	2335	26	2432	7	4.0
P	24	21.1	567	374	0.66	0.443	2.181	0.031	0.16664	0.00075	2.190	0.031	0.16270	0.00099	2424	29	2484	10	2.4
D	7	7.1	2163	598	0.28	0.075	3.116	0.038	0.12302	0.00041	3.119	0.038	0.12236	0.00044	1793	19	1991	6	9.9
D	3	3.1	939	1784	1.90	0.101	2.568	0.023	0.14557	0.00118	2.571	0.023	0.14467	0.00120	2118	16	2284	14	7.2
D	28	25.1	1011	416	0.41	0.203	2.656	0.025	0.14690	0.00129	2.662	0.025	0.14510	0.00149	2056	17	2289	18	10.2
D	30	27.1	705	783	1.11	0.075	2.690	0.038	0.15614	0.00076	2.692	0.038	0.15547	0.00080	2037	25	2407	9	15.4
D	14	14.1	458	257	0.56	0.038	2.746	0.028	0.16098	0.00088	2.748	0.028	0.16064	0.00090	2001	18	2462	9	18.7
D	18	15.1	537	214	0.40	0.085	2.479	0.025	0.16745	0.00084	2.481	0.025	0.16669	0.00089	2183	19	2525	9	13.5
D	27	24.1	459	299	0.65	0.032	3.095	0.033	0.17009	0.00232	3.096	0.033	0.16980	0.00233	1804	17	2556	23	29.4
D	5	5.1	685	66	0.10	0.014	2.256	0.030	0.17128	0.00136	2.256	0.030	0.17116	0.00136	2365	27	2569	13	7.9
D	9	9.1	330	203	0.61	0.036	2.426	0.028	0.17269	0.00187	2.426	0.028	0.17237	0.00189	2225	22	2581	18	13.8
D	16	15.1	278	46	0.17	0.658	2.287	0.029	0.18645	0.00131	2.302	0.029	0.18058	0.00172	2326	25	2658	16	12.5
D	13	13.1	208	62	0.30	0.210	2.442	0.033	0.18389	0.00149	2.447	0.034	0.18202	0.00165	2209	26	2671	15	17.3
D	20	17.1	199	49	0.24	0.048	2.098	0.028	0.18455	0.00135	2.099	0.028	0.18412	0.00138	2512	28	2690	12	6.6

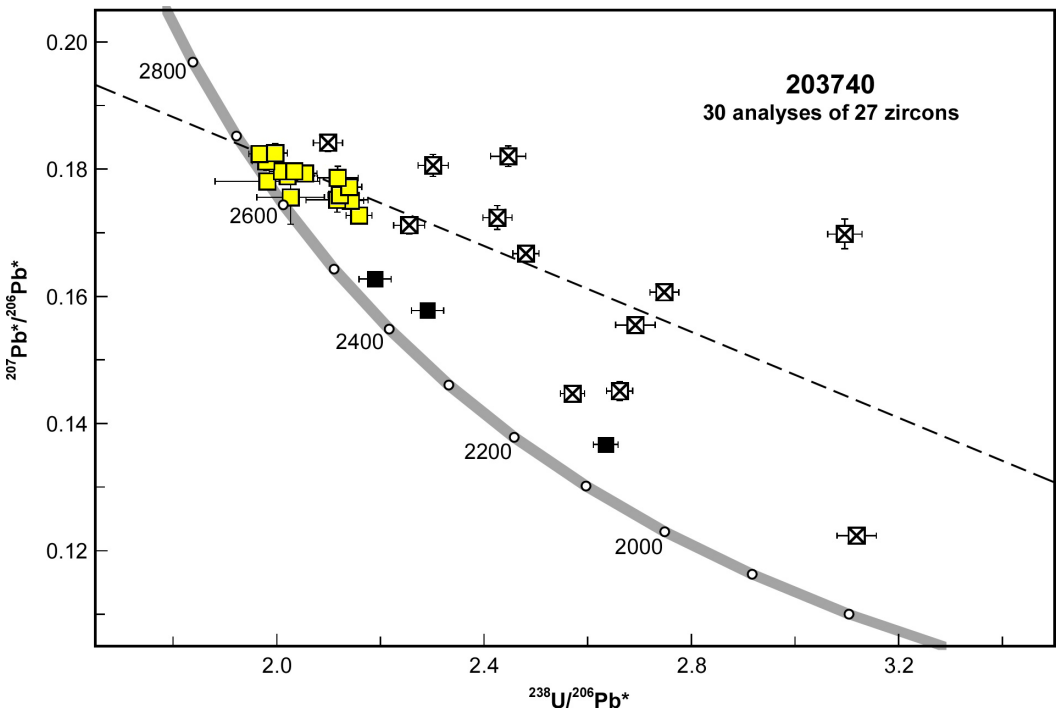

Figure 3. **U–Pb analytical data for sample 203740: porphyritic metasyenite, Wallaby mine. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >6%). The dashed line indicates a discordia regression through data in Group I**

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Twelve analyses are >6% discordant. The dates obtained from these 12 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 17 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses of 14 zircons (Table 1), which yield a discordia with intercepts at 2688 ± 27 and c. 1400 Ma (MSWD = 0.58).

Group P comprises three analyses of three zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2484–2185 Ma.

The date of 2688 ± 27 Ma for the 15 analyses in Group I is interpreted as the age of igneous crystallization. An alternative estimate of the igneous crystallization age is provided by the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2649 ± 12 Ma (MSWD = 2.0) for the oldest 10 analyses in Group I. The dates of 2484–2185 Ma for the three analyses in Group P are interpreted to reflect loss of radiogenic Pb.

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Recommended reference for this publication

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